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Deep Learning-Based Estimation of Petrophysical Properties for Unconventional Shales from SEM Images

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Abstract

The use of machine learning algorithms to process images has significantly enhanced our ability to extract meaningful information in diverse areas such as weather forecasting, medical imaging, satellite image analyses, and driverless vehicle navigation. Borrowing on these technologies, we demonstrate the use of deep-learning features acquired from an image classifier to predict petrophysical properties in unconventional shales. By directly analyzing images, that can easily be acquired from drill cuttings or even rock chips in highly friable formations which make routine core analyses challenging, we are able to provide very rapid turnover at a fraction of the cost of lab-based methods that are more rigorous, but also more resource-intensive.

Our dataset consists of over 8000 images from 22 shale plays across North and South America and Europe. We first construct a convolutional neural network-based (CNN) classifier trained to recognize the source play associated with each image. In this work, we use the well-known ResNet50 architecture. We then extract the deep learning features from the model after training, and relate these to various petrophysical properties such as porosity, mineralogy and elastic properties such as Young's Modulus.

We obtain a very high degree of precision, recall and accuracy over all 22 classes with the CNN-based classifier. This demonstrates that the CNN is able to capitalize and exploit microstructural differences between each of the 22 plays to identify the source of an image. However, the misclassifications that did occur point to similarities in the microstructure of two or more plays. In situations such as these, the different plays can be considered to be digital geologic analogs that can provide useful information for completion and well design as well as the planning of drilling operations. By extracting the deep-learning features from this trained CNN-based network, we were able to show remarkable predictive capability for petrophysical properties such as Young's Modulus, porosity and mineralogy.

Prior approaches to predict petrophysical properties from images have relied on image segmentation (identifying the pixels associated with grains, pores and other features in an image) and then relating them to petrophysical properties. Image segmentation, for the complex microstructure associated with the SEM images can take upwards of two-three weeks per image, making the traditional approach prohibitively

expensive. In this work, by using higher-level deep learning features from a trained classifier, we circumvent this problem and present a more robust approach to image-based estimation of petrophysical properties.

Introduction

Convolutional Neural Networks (CNNs) are routinely employed across a wide range of applications from medical image analysis to autonomous driving. Their unique ability to automatically learn spatial hierarchies as well as their translation invariance with respect to images makes them a powerful tool in computer vision applications.

CNNs, such as AlexNet (Krizhevsky et al., 2012), VGG (Simonyan & Zisserman, 2014), and ResNet (He et al., 2015), have proven to be highly effective in object detection. By stacking convolutional layers, CNNs can progressively learn more complex features of the image, starting from simple edges to more complex objects like faces, animals, or buildings or in the case of rocks, features such as pores, different grains and fractures. However, training a CNN necessitates large volumes of labeled data and computational resources, particularly GPUs for parallel computation.

Once a CNN is trained to classify images, its learned features can be leveraged for estimating various image properties. Such examples include detecting and measuring the size of tumors in medical scans (Jiao et al. 2019), or determining the roughness of metal surfaces by analyzing the texture patterns in the image (Zhang et al. 2019) estimating the age of a person in an image, which can be useful for personalized marketing (Geng et al. 2013).

The estimation of petrophysical properties, such as porosity, permeability, and mineralogy as well as elastic properties such as Young's modulus, compressibility and Poisson's ratio, from rock samples plays a critical role in reservoir characterization. Traditionally, these are estimated through core analysis, well logging, and the use of correlations, all of which require significant time, resources, and expertise. However, there has been a shift toward using deep learning (DL) models to predict these properties from high-resolution 3D imaging from X-ray tomography (CT scans) and focused ion beam scanning electron microscopy (FIB-SEM).

There have been several applications of deep learning for geological image classification tasks, including shale FIB-SEM images (Knaup et al., 2019; H. Zhang et al., 2021; Knaup et al., 2022), core samples (Wang et al., 2019; Kathrada and Adillah, 2019; B. Zhang et al., 2021; Carpenter, 2024), and thin sections (Mosser et al., 2022; Tolstaya et al., 2023). Other applications of CNNs have focused on capturing pore structures to predict porosity and permeability values. Yu et al. (2020) trained CNNs on X-ray CT images of rocks, with known porosity and permeability values, to predict these properties for new, unseen images. A study by Alharthi et al. (2021) uses a deep learning framework that analyzes both well-log data and core sample images to predict petrophysical properties such as porosity and permeability. Li et al. (2019) applied deep learning to predict the geomechanical properties of rocks using well log data and were able to predict the elastic moduli of different lithologies. Such studies demonstrate the potential for rapid, non-invasive image-derived petrophysical property estimation that are especially valuable in cases where physical core samples are difficult or expensive to acquire.

In two previous studies (Yang et al. 2023 and Maryada et al. 2025), we showed how textural attributes can be used to predict the Young's modulus for fine-grained rocks. We also demonstrated that, contrary to what is generally assumed, the use of mineralogy as well as textural attributes to predict the Young's modulus did not provide any additional benefit over using just the textural attributes alone. The conclusion from the study was that texture, captured from the SEM images, is sufficient to capture the mineralogy (through variations in grayscale levels), but more importantly, provides a gross estimation of the spatial arrangement and heterogeneity of the different minerals. Our understanding is that texture alone captures the mineralogical variations within the samples without the need for explicitly estimating/measuring mineralogy.

The paper is organized as follows: We first describe our methodology to construct a grayscale FIB-SEM image classifier. Our images were acquired from 22 different unconventional shale plays, and we classify images according to the play or source. This is followed by a discussion of the image features captured by the classifier. Finally, we demonstrate how we can use features to predict the Young's modulus and porosity directly from FIB-SEM images.

Data Available

The dataset used in this study comprises 8000 FIB-SEM grayscale BSE images collected from 22 shale plays across North and South America and Europe. Each image was standardized to 8-bit depth, resized to 224×224 pixels and rescaled to a resolution of 50 nm per pixel, resulting in over 22,000 images. The dataset was split into 80% for training and 20% for testing and prediction. To further enhance consistency, all images were normalized based on their mean and standard deviation.

It is important to mention that the images for each of the 22 plays were acquired from a few adjacent wells. Given that a play can exhibit varying characteristics at different locations and depths, our classification by source or play is simply to gauge if there is microstructural uniqueness in images from a specific source or alternatively, if two sources share microstructural commonalities.

We then select samples from 14 of the 22 unconventional shale plays for which we possess both porosity and Young's Modulus data in our study. These are listed in [Table 1](#). The Young's Modulus measurements were acquired using nanoindentation which is suitable for the fine-grained rocks (Acosta et al. 2020). The measurement involves a load applied to the sample using a nanoindenter diamond tip. The load is progressively increased to a specific limit and then held for a specific time before release. The elastic modulus and hardness are estimated during the unloading phase. To avoid measurements that are specific to any large-grained mineral, if any, in the sample, the average modulus is estimated from multiple indentations. The porosity measurements available to this study were made using Nuclear Magnetic Resonance (NMR) and Helium Porosity Pycnometry (HPP) and represents both the connected and unconnected porosity in the sample.

Table 1—The list of fourteen plays used in this study for which measured values of porosity and Young's Modulus were available.

Alum	Kimmeridge
Barnett	La Luna
Collingwood	Marcellus
Duvernay	Niobrara
Eagle Ford Gas	Osage
Eagle Ford Oil	Utica
Haynesville	Vaca Muerta Gas

Methodology

In a previous study (Knaup et al. 2023), we used a variant of AlexNet ([Krizhevsky et al. 2012](#)) to classify images from various unconventional plays. The goal was to assess if deep learning algorithms can recognize and classify FIB-SEM images and correctly label the source of the image. More recently, in [Maryada et al. \(2025\)](#), we performed a similar classification task using ResNet50 architecture. However, in the more recent work, our goals were different, and we were attempting to extract deep learning features and use textural information to relate to various petrophysical properties. In this paper, we solely focus on the deep learning features acquired from a trained ResNet50 classifier.

Deep learning networks for classification or any other computer vision task rely on successive sequences of convolution and pooling. Although individual network architectures might vary, the various trained models are associated with a key commonality. That is, the output derived following the last step of convolution and pooling is expected to contain features that are relevant to the classification problem. In the context of SEM image classification, these features are expected to contain information relevant to the unique microstructure associated with different images from different plays. In a trained deep learning-based classification network, the processed output will reflect the microstructural uniqueness embedded in an image.

In this study, we employed the ResNet-50 model to classify the source of the images used in this study. The reader is referred to Zhang et al. (2022) for more information. While the accuracy of the model was high (greater than 97%), we also ensured that the rate of misclassification was low. Fig. 1 shows the confusion matrix associated with the trained model indicating a low rate of misclassification.

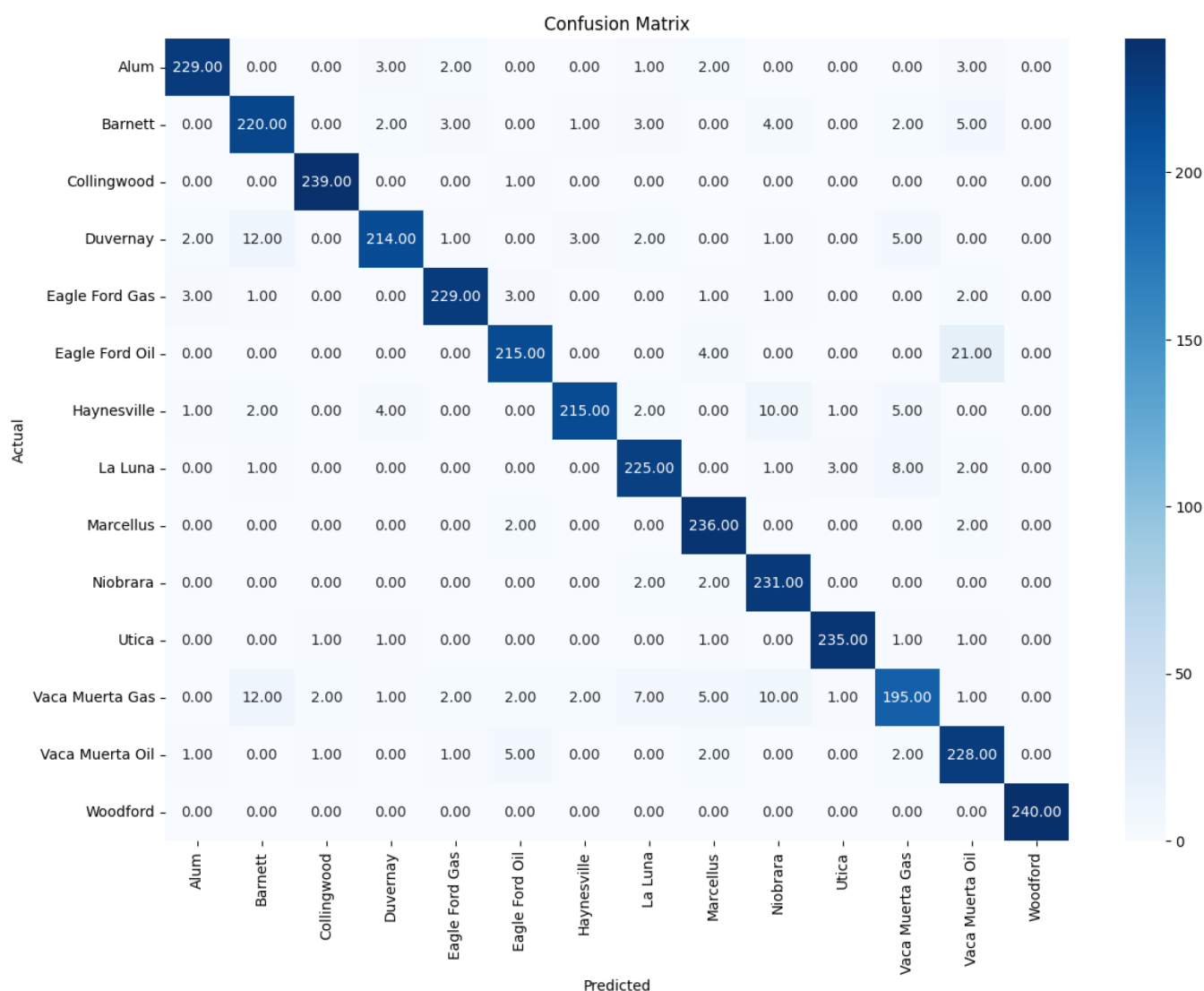


Figure 1—The confusion matrix associated with the trained ResNet50 model operating on test images. The model has a 97% accuracy and shows a low rate of misclassification. In a previous paper (Zhang et al. 2022) we show an approach to find microstructural analogs when misclassification is persistent among images associated with two plays.

The ResNet-50 model, when used as a feature extractor, produces a feature vector of length 2048. This vector contains the learned representations from the last global average pooling layer before the final

fully connected layer of the ResNet-50 architecture. These features are often considered high-level and semantically meaningful, making them useful for various downstream tasks such as image classification, object detection, and image retrieval. In this paper, we follow the process outlined in Fig. 2.

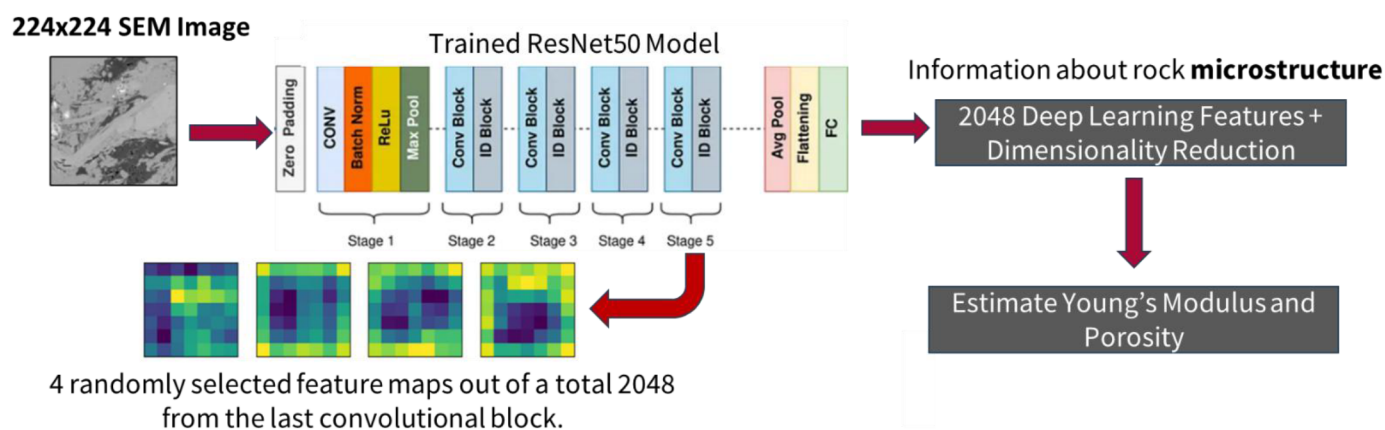


Figure 2—The overall workflow followed in this work. Images fed to a trained ResNet50 classifier provide 2048 7×7 feature maps in the last layer. Taking the average of each feature map and selecting the most relevant features allows us to use image-derived data to predict Young's Modulus and porosity values in this paper. Figure adapted from Maryada et al. (2025)

As an example, consider Fig. 3 containing a randomly selected FIB-SEM image from a specific play. As this image progresses through a trained ResNet50 network, we can extract feature maps from each successive convolutional layer to illustrate the propagation of information from one layer to a succeeding layer. This process is illustrated in Fig. 4. Fig. 4(a) shows four randomly selected feature maps obtained after the second convolution layer from the trained ResNet50 model. There are two observations that can be made. The feature maps following two stages of convolution and pooling are slightly different from one another yet retain some semblance to the original image in Fig. 1, albeit at a lower resolution. Likewise, Fig 4(b) shows four other randomly selected feature maps from the third convolution layer, (c) from the fourth, (d) from the fifth and last convolution layer at a 7×7 resolution. As the image progresses through successive steps of convolution, the feature maps obtained are of a lower resolution, till finally in the last layer, there is little resemblance to the original image.

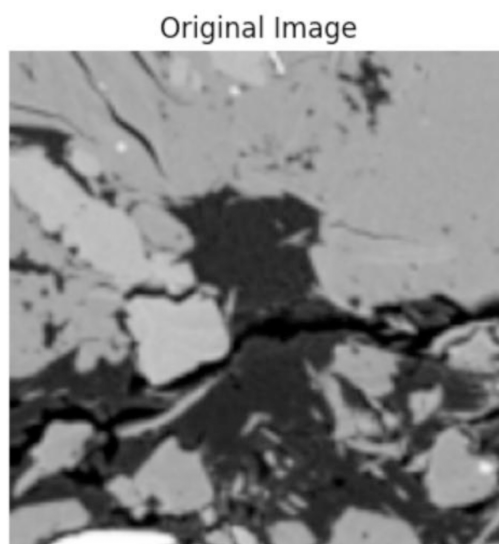


Figure 3—The figure shows a randomly selected grayscale FIB-SEM image acquired from a specific play.

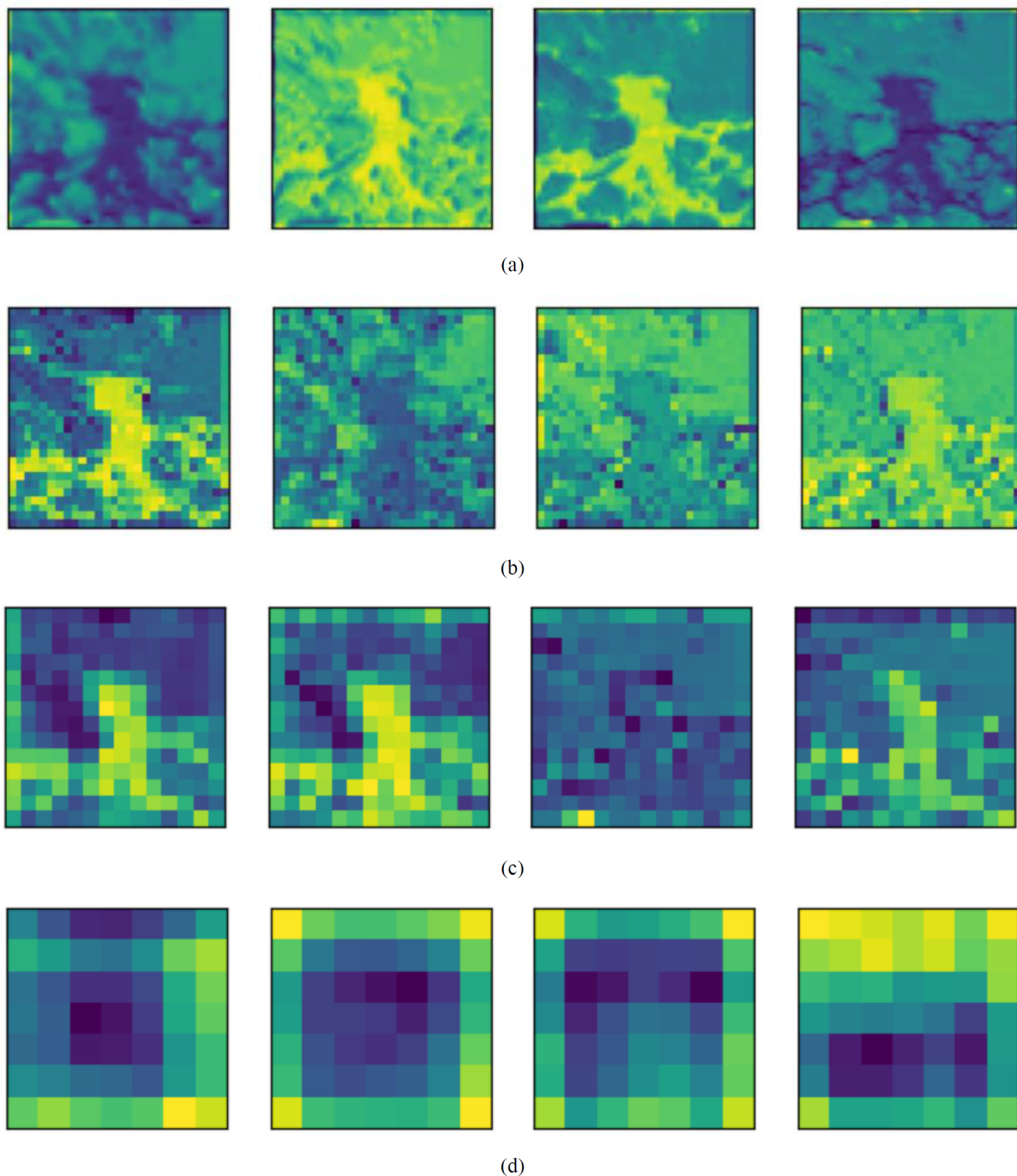


Figure 4—The figure above illustrates feature maps acquired from a trained image classifier. The trained classifier is based on the ResNet50 architecture and is trained to label the source of an image from among 22 different unconventional plays. (a) shows four randomly selected feature maps obtained after the second convolution layer. There are two observations that can be made. The feature maps following two stages of convolution and pooling are slightly different from one another yet retain some semblance to the original image, albeit at a lower resolution. Likewise, (b) shows four other randomly selected feature maps from the third convolution layer, (c) from the fourth, (d) from the fifth and last convolution layer at a 7×7 resolution. As the image progresses through successive steps of convolution, the feature maps are at a lower resolution, till finally in the last layer, there is little resemblance to the original image with 224×224 pixels.

Nevertheless, the feature maps obtained after the fifth convolutional layer should contain microstructural information relevant to the classification of the original image in Fig. 1. It is this information that we seek to harness. Our goal is to relate the information content embedded in the 2048 final 7×7 feature maps of the ResNet50 to relate to petrophysical properties. Because the microstructural information is unique to each image, it is also likely to be indicative of petrophysical and geomechanical properties of the sample. Our work in this paper demonstrates relating these properties to the deep learning feature maps from the last layer of the trained ResNet50 model.

Fig. 5a shows another randomly selected FIB-SEM image in our dataset. When comparing four randomly selected feature maps extracted from the image in Fig. 5a and shown in Fig. 5b for Layer 4 and Fig. 5c for Layer 5, we observe that as we progress deeper into the network, the information embedded in the original image is extracted to lower resolutions and finally in the last layer, there is little resemblance to the original image, analogous to the observations made in Fig. 4. However, and more importantly, the 7×7 feature maps in Fig. 5c associated with the fifth convolutional layer are significantly different from those associated with a different image as shown in Fig. 4d. This qualitatively illustrates the microstructural uniqueness associated with the two original images and captured by the deep learning feature maps that enables the quantification of petrophysical and geomechanical properties from images.

Finally, we use a decision tree regression model paired with LOOCV (Leave-One-Out-Cross-Validation) for validation. Our dataset consists of 14 data points, and this prevents the use of traditional train-test splits or k-fold cross-validation. We chose to use a non-parametric decision tree regression model because of the smaller size of our dataset. We then report the root-mean-squared-error (RMSE) and R^2 values obtained during LOOCV to demonstrate that our model can generalize to new data and new images.

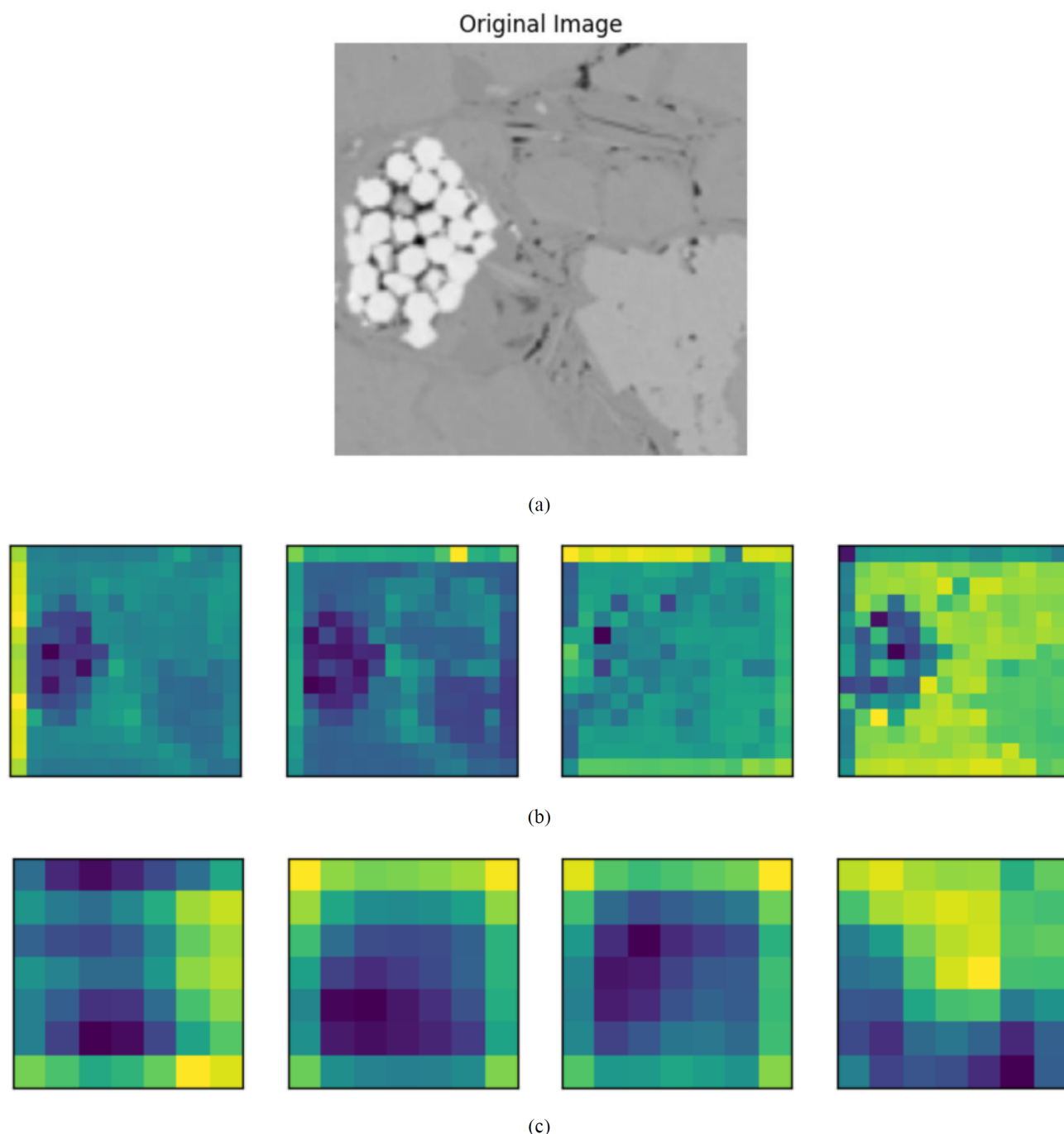


Figure 5—The figure in (a) shows a different randomly selected grayscale FIB-SEM image from within our dataset of images. (b) shows four randomly selected feature maps from the fourth convolutional layer and (c) shows the corresponding maps obtained from the fifth convolutional layer. The differences between these images and those in [Fig. 4](#) illustrate that microstructural uniqueness can be harnessed to provide petrophysical and geomechanical properties directly from images.

Results and Detailed Workflow for Subsequent Use

After extracting the 2048 feature maps following the last convolutional layer in the trained ResNet50 model and converting them to a vector of length 2048 using global average pooling, we then select the 40 most relevant features. These are then related to both the Young's Modulus and the porosity associated with the samples using a non-parametric decision tree regression model.

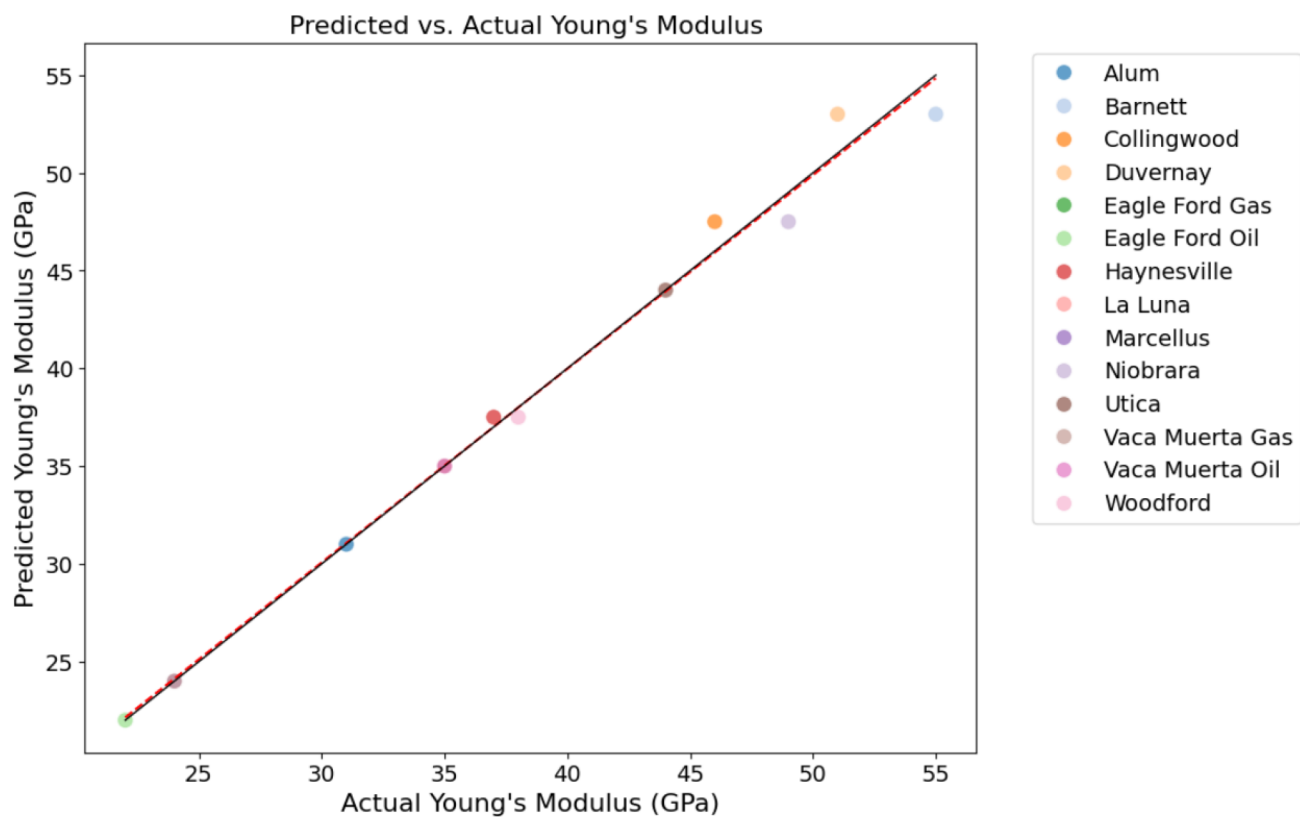


Figure 6—The plot above shows the predicted Young's Modulus versus the actual Young's Modulus during leave-one-out-crossvalidation (LOOCV). The RMSE for the regression model was under 2.1 GPa.

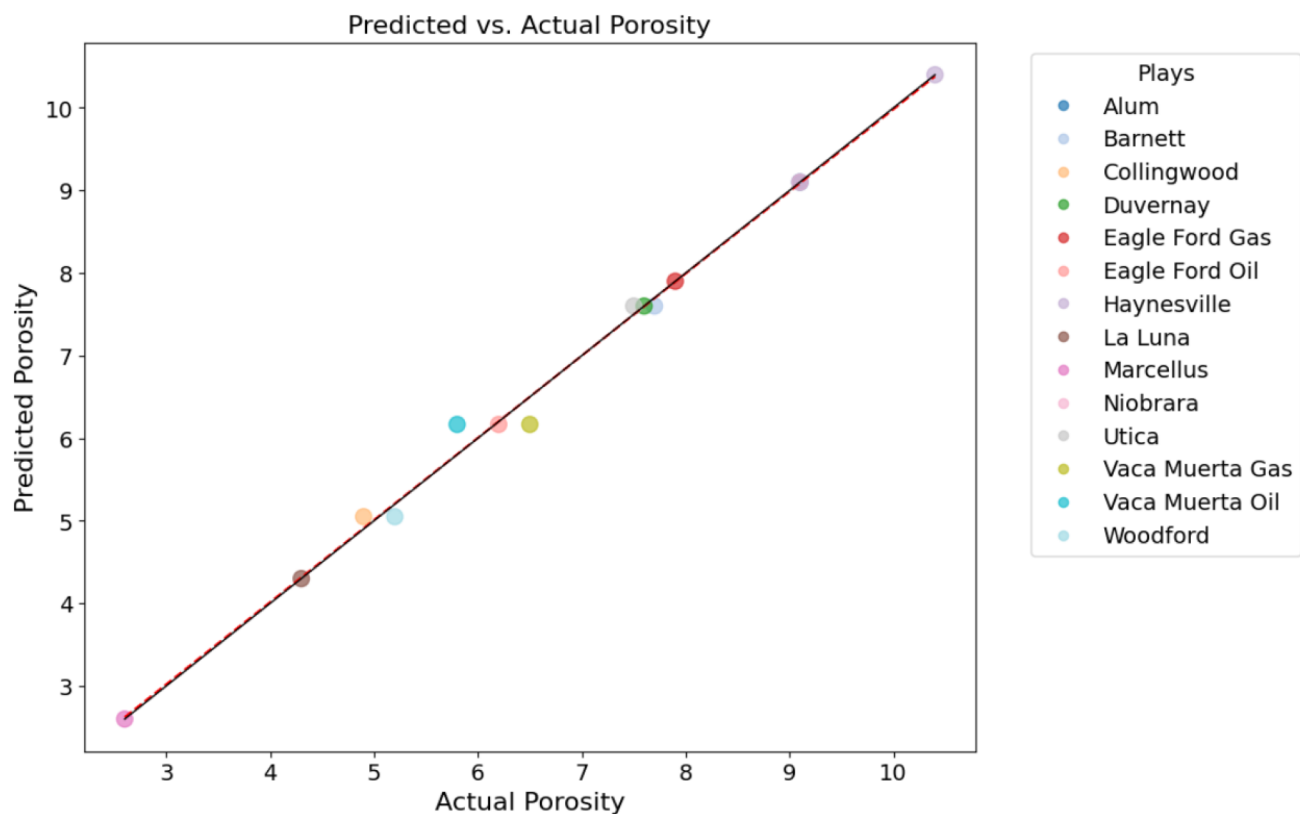


Figure 7—The plot above shows the predicted porosity values versus the actual during leave-one-out-crossvalidation (LOOCV). The RMSE for the regression model was under 0.2 p.u.

Figs. 6 and 7 above show the predicted versus actual Young's Modulus and porosity values respectively. In both cases, we can observe a remarkable agreement between the predicted and actual values for all plays, even though the porosity values and Young's moduli values span very large ranges. Overall, this indicates the promise of information embedded in FIB-SEM data in predicting petrophysical properties. Consequently, this allows for near real-time prediction of these properties using, for example, drill cuttings with the appropriate depth registration. While drill cuttings have traditionally been used in petrographic analyses and to identify gas/oil shows, the application shown in this paper demonstrates how petrographic analyses can be extended to even predict petrophysical and geomechanical properties.

In principle, applying this methodology to drill cuttings would require FIB-SEM imaging of several (#10) samples acquired from the same depth interval. These images would then need to be processed through the trained ResNet50 model to acquire the deep learning feature vectors. These feature vectors are averaged over each of the samples and finally, the trained decision tree model is used to predict either Young's modulus or porosity using the averaged deep learning vector as inputs. It is important to mention that the averaging process is a necessary aspect of this workflow. Petrophysical/geomechanical properties are acquired using samples that are typically over an inch or more in length. For the image data to be representative of the information contained in larger samples, it is necessary to acquire several images, followed by averaging of their feature maps/vectors.

Conclusions

In this paper, we demonstrate a promising approach that leverages microstructural information distilled from FIB-SEM images of unconventional shales using a trained image classifier to predict petrophysical and geomechanical properties. The trained classifier in this paper uses the ResNet50 architecture, but any CNN-based deep learning model may be used instead. The following conclusions can be drawn from this study:

1. We construct a trained classifier based on the ResNet50 architecture which was successfully able to label over 8000 images according to their source.
2. The methodology described shows excellent predictive capability for both Young's modulus and porosity values using only image data processed through a trained classifier.
3. For practical application, it will be necessary to acquire several (#10) images over a representative volume to estimate any petrophysical or geomechanical property. In the case of drill cuttings, it may be necessary to image 8-10 samples for robustness in predictability. This will require approximately 24 hours to process.

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